

Optimal Acoustic Parameter Subsets for Dimension-Specific Voice Quality Prediction

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SUMMARY: Objectives. To determine the minimal set of acoustic parameters that optimally predicts each major perceptual voice quality dimension and to quantify the redundancy among commonly measured acoustic voice parameters.

Study design. Secondary analysis of a publicly available voice database.

Methods. Fourteen acoustic parameters were extracted from sustained vowels of 284 speakers in the Perceptual Voice Qualities Database (PVQD). Four right-skewed parameters (jitter, shimmer, shimmer dB, and period standard deviation) were log-transformed to linearize their relationship with perceptual ratings. QR factorization with column pivoting characterized the redundancy structure among parameters. Orthogonal Matching Pursuit (OMP) ranked parameters by incremental predictive value for each perceptual dimension (CAPE-V Severity, Breathiness, Roughness, and Strain; GRBAS Grade, Breathiness, and Roughness). The optimal number of parameters was determined by convergence of adjusted R^2 , Bayesian Information Criterion, and cross-validated prediction error. Classification performance against dichotomized perceptual ratings was evaluated using ROC analysis with 10-fold stratified cross-validation.

Results. The 14-parameter acoustic space had an effective dimensionality of approximately 11, with slope contributing negligible independent information. Log transformation improved predictive accuracy for all dimensions, except breathiness, with gains that were modest for overall severity and substantial for roughness. Optimal parameter subsets varied across perceptual dimensions: overall severity required six parameters (log shimmer dB, GNE, Hno-6000, log PSD, F0, and HNR-D; $r_s = 0.77$, AUC = 0.870), breathiness required three (CPPS, GNE, and Hno-6000; $r_s = 0.73$, AUC = 0.865), roughness required 5 (log shimmer dB, H1-H2, GNE, log PSD, and log jitter; $r_s = 0.71$, AUC = 0.830), and strain required six (HNR, F0, log PSD, tilt, H1-H2, and GNE; $r_s = 0.66$, AUC = 0.790). The severity subset yielded a higher area under the curve (AUC) than the Acoustic Voice Quality Index (AVQI: six parameters, AUC = 0.827) and the Cepstral Spectral Index of Dysphonia (CSID: AUC = 0.787). This advantage was obtained from sustained vowels alone, whereas AVQI and CSID were scored on connected speech in addition to the vowel; the comparison is therefore task-asymmetric and favors the established indices in speech material. Notably, log shimmer dB displaced smoothed cepstral peak prominence (CPPS) as the most important predictor of overall severity, suggesting that the widely reported primacy of CPPS may partly reflect its distributional advantage in linear models.

Conclusions. Different perceptual dimensions of voice quality are optimally predicted by distinct subsets of acoustic parameters. Log-transforming right-skewed perturbation measures improves linear prediction incrementally for overall severity and substantially for roughness. Using sustained vowels alone, data-driven parameter selection matches or exceeds the classification performance of established multiparametric indices scored on more speech material, and identifies candidate parameter sets for roughness and strain—dimensions lacking published composite indices. External validation and extension to connected speech remain necessary before clinical adoption.

Key Words: Acoustic voice analysis—Feature selection—QR factorization—Voice quality—PVQD—Multiparametric indices.

INTRODUCTION

Clinical acoustic voice analysis involves measuring multiple parameters that quantify different aspects of the voice signal: periodicity (jitter, shimmer), noise content (harmonics-to-noise ratio, glottal-to-noise excitation ratio),

spectral characteristics (cepstral peak prominence, spectral slope), and source properties (fundamental frequency, H1–H2 amplitude difference). A persistent challenge is that many of these parameters are substantially intercorrelated, creating redundancy that complicates clinical interpretation and index design.^{1,2}

This redundancy has been addressed in two ways. First, expert-designed multiparametric indices combine selected parameters into single scores: the Acoustic Voice Quality Index (AVQI)³ uses six parameters for overall dysphonia severity, the Acoustic Breathiness Index (ABI)⁴ uses nine parameters for breathiness, and the Cepstral Spectral Index of Dysphonia (CSID)⁵ uses cepstral and spectral measures for dysphonia severity. These indices were developed

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through stepwise multiple regression, where parameters were selected based on their incremental contribution to predicting perceptual ratings in specific development samples. Second, various studies have applied feature selection methods—including principal component analysis (PCA),⁶ correlation thresholding,⁷ variance-based filtering,⁸ and machine learning approaches⁹—to reduce the parameter space.

However, several questions remain open. First, the degree of redundancy among acoustic parameters has not been formally quantified using rank-revealing matrix decompositions, which provide a principled measure of the effective dimensionality of the parameter space. Second, existing multiparametric indices were designed for specific perceptual dimensions (severity, breathiness), while other clinically important dimensions—notably roughness and strain—lack dedicated indices. Third, it is unknown whether the optimal parameter subset varies across perceptual dimensions or whether a common core of parameters suffices for all dimensions. Finally, no study has systematically compared dimension-specific parameter rankings derived from the same database and methodology.

QR factorization with column pivoting is a classical technique from numerical linear algebra that greedily selects the most linearly independent columns of a matrix, producing a rank-revealing decomposition.^{10,11} It has been applied to feature selection in speaker identification¹² and biomedical signal classification,¹³ but has not been used in clinical voice quality assessment. Orthogonal Matching Pursuit (OMP) extends this approach by incorporating a prediction target: it iteratively selects the feature most correlated with the current prediction residual, refits the model, and updates the residual.¹⁴ OMP is closely related to forward stepwise regression—the method originally used to develop both AVQI³ and ABI⁴—but operates on standardized, normalized predictors and uses residual correlations rather than p values for feature selection, making it less sensitive to collinearity and sample-specific distributional properties. Combined, QR factorization (for characterizing redundancy without reference to any perceptual target) and OMP (for target-specific ranking) provide a transparent, deterministic framework for parameter selection.

The objectives of this study were: (1) to quantify the redundancy structure among 14 acoustic parameters measured from sustained vowels, (2) to determine the rank-ordered importance of these parameters for predicting each major perceptual voice quality dimension, (3) to identify the optimal number of parameters for each dimension using multiple model selection criteria, and (4) to evaluate the classification performance of the resulting dimension-specific indices relative to established multiparametric indices, within the sustained-vowel task. Sustained vowels were used rather than connected speech because they isolate voice source characteristics (glottal vibration, noise generation) from articulatory and linguistic influences, and yield language-independent acoustic measurements.

METHODS

Database

Voice recordings were obtained from the Perceptual Voice Qualities Database (PVQD),¹⁵ a publicly available collection of 296 speakers with and without voice disorders. Each recording was rated by at least three experienced voice clinicians using both the Consensus Auditory-Perceptual Evaluation of Voice (CAPE-V)¹⁶ and the Grade, Roughness, Breathiness, Asthenia, Strain (GRBAS) scale.¹⁷ Recordings were obtained using a head-mounted condenser microphone at a 6 cm mouth-to-microphone distance, with 16-bit quantization at either 44.1 kHz or 25 kHz sampling rate.

Audio preprocessing

Sustained /a/ vowel segments were extracted from each recording. Leading and trailing silence was trimmed using an intensity threshold of -30 dB relative to maximum. No amplitude normalization or resampling was applied; all files were processed at their native sampling rate. After preprocessing, files with voiced duration under 0.5 seconds were flagged but retained in the analysis.

The sample was assembled as follows. Of the 296 recordings in the database, 290 yielded usable sustained-vowel segments; six were excluded for missing or corrupted audio.¹⁸ Matching against the perceptual ratings removed two further speakers for whom no ratings were available, giving 288 merged records. Of these, four were excluded because one or more acoustic parameters could not be computed (one recording did not permit estimation of shimmer in dB; three did not permit estimation of the LTAS slope and tilt). Complete-case analysis therefore yielded a final sample of $N = 284$ speakers. This same sample is used for the QR factorization, the OMP rankings, all regression models, and all classification analyses; no perceptual dimension is fitted on a different subset.

Sampling-rate heterogeneity. The PVQD is not distributed at the sampling rate stated in its documentation. The database overview reports a single 48 kHz rate for all recordings,¹⁹ but direct inspection of the distributed audio files shows otherwise: of the 290 extracted vowel segments, 219 are at 44.1 kHz and 71 at 25 kHz, and none at 48 kHz. Because Praat computes HNR, CPPS, GNE, jitter, and shimmer over the full spectrum up to the Nyquist frequency, a systematic difference in acquisition bandwidth could in principle bias these measures and, in turn, distort data-driven parameter selection. This was tested directly. Every 44.1-kHz recording was resampled to 25 kHz with anti-alias filtering and all 14 parameters were re-extracted, so that each recording could be compared with itself under the two bandwidths. The resulting shifts were negligible relative to between-speaker variation: the largest standardized effect was $d = 0.10$ (Hno-6000), and all others were $d \leq 0.07$, with HNR and CPPS changing by less than 0.05 dB. Acquisition bandwidth is therefore not a material source of

TABLE 1.
Acoustic Parameters Extracted From Sustained Vowels

Parameter	Unit	Category	Description	Index
F0 mean	Hz	Source	Mean fundamental frequency	—
Jitter local	%	Periodicity	Cycle-to-cycle F0 perturbation	ABI
Shimmer local	%	Periodicity	Cycle-to-cycle amplitude perturbation	AVQI, ABI
Shimmer local	dB	Periodicity	Shimmer in dB	AVQI, ABI
HNR	dB	Noise	Harmonics-to-noise ratio (Praat standard)	AVQI
CPPS	dB	Cepstral	Smoothed cepstral peak prominence	AVQI, ABI, and CSID
Slope	dB	Spectral	LTAS slope (0-1 vs 1-10 kHz)	AVQI
Tilt	dB	Spectral	LTAS trend line slope	AVQI
Alpha ratio	dB	Spectral	Energy ratio (0-1 vs 1-5 kHz)	—
GNE	—	Noise	Glottal-to-noise excitation (max at 4500 Hz)	ABI
Hno-6000	—	Noise	High-frequency noise ratio (0-6 vs 6-10 kHz)	ABI
HNR-D	dB	Noise	HNR, Dejonckere variant	ABI
H1-H2	dB	Source	First-second harmonic amplitude difference	ABI
PSD*	s	Periodicity	Period standard deviation	ABI

Notes: The *Index* column gives membership in the established multiparametric indices against which the study benchmarks. Twelve of the 14 parameters are components of AVQI³ or ABI⁴; mean F_0 and the alpha ratio are additions, included to represent source characteristics and spectral balance respectively; CSID⁵ contributes no parameter beyond CPPS.

* PSD here denotes *period standard deviation*, the frequency short-term perturbation measure used in ABI,⁴ and not power spectral density. The abbreviation is used in both senses in the voice literature.

bias for these parameters in this database, and the full sample was retained without restriction or resampling.

Acoustic parameters

Fourteen acoustic parameters were extracted from each pre-processed sustained vowel using *Parselmouth* version 0.4.7²⁰ (embedding *Praat* 6.1.38). The parameter set (Table 1) was designed to encompass the full range of acoustic dimensions relevant to voice quality perception, namely periodicity measures, noise-related measures, cepstral measures, spectral shape measures, and source characteristics. It comprises *all acoustic components of AVQI³ (six parameters) and ABI⁴ (nine parameters)*—12 distinct parameters in total, since CPPS, shimmer (%), and shimmer (dB) are common to both—supplemented by two additions, mean F_0 and the alpha ratio, which represent source characteristics and spectral balance and are not measured directly by either index. Table 1 gives the membership of each parameter.

This criterion matters for the interpretation of the comparisons that follow. Because the data-driven subsets are drawn from a feature space that *contains* every component of AVQI and ABI, any difference in predictive performance relative to those two indices is attributable to parameter selection and transformation rather than to the introduction of new features. The same does not hold for CSID,⁵ whose low-to-high spectral energy ratio component—computed over 0–4 and 4–8 kHz—is not among the 14 parameters and could therefore never be selected here. The comparison with CSID is consequently not one of selection within a shared feature space, and is interpreted with corresponding caution.

Three classes of measure were excluded by design. Parameters computable only from connected speech were

excluded because the study isolates the sustained-vowel task (*Task Dependence and Ecological Validity*). Nonlinear dynamic measures (eg, recurrence period density entropy, detrended fluctuation analysis, and correlation dimension) and machine learning representations (eg, mel-frequency cepstral coefficients) were excluded because they are not interpretable as individual clinical parameters, and their inclusion would defeat the aim of producing transparent, dimension-specific indices.

Parameter transformation

Four parameters—jitter (%), shimmer (%), shimmer (dB), and PSD—were log-transformed prior to all subsequent analyses to linearize their relationship with perceptual ratings. This transformation is motivated by the strongly right-skewed distributions of these parameters (coefficient of variation > 0.9), which compress discriminative information for the majority of voices into a narrow range while extending a long tail for severely disordered voices. Log transformation has been applied to jitter and shimmer in previous voice research to stabilize measurement variance,²¹ but its effect on perceptual prediction accuracy does not appear to have been previously reported. The remaining 10 parameters were used untransformed: HNR, H1-H2, slope, tilt, and alpha ratio contain negative values (precluding logarithmic transformation), while F_0 , CPPS, GNE, Hno-6000, and HNR-D have approximately symmetric distributions.

Perceptual ratings

Perceptual ratings were the mean across all available raters for each speaker, as provided in the PVQD dataset. Seven perceptual targets were analyzed: CAPE-V Overall

TABLE 2.
Reliability of the Perceptual Reference Standard
(Intraclass Correlations, Averages Used as Ratings), as
Reported by the Database Authors¹⁵

Perceptual Dimension	Inter-rater ICC	Intra-rater ICC
CAPE-V Overall Severity	0.918	0.943
CAPE-V Breathiness	0.827	0.911
CAPE-V Strain	0.829	0.908
CAPE-V Roughness	0.789	0.896
GRBAS Grade	0.911	0.905
GRBAS Breathiness	0.844	0.884
GRBAS Roughness	0.787	0.846

Severity, Roughness, Breathiness, and Strain; GRBAS Grade, Roughness, and Breathiness.

In the PVQD, each listener rated approximately 50 files, and most files were rated by three experienced voice clinicians (some by four), all with at least three years of clinical experience.¹⁵ Raters and speakers are therefore not fully crossed, and individual rater effects cannot be modeled analytically; the published mean ratings were used as the perceptual reference standard, consistent with other PVQD-based studies.⁶ Inter-rater reliability for the averaged ratings was computed by the database authors using intraclass correlations and is reproduced in Table 2; intra-rater reliability, assessed by rating each file twice, was likewise high (overall ICC = 0.912 for CAPE-V and 0.889 for GRBAS).¹⁵

Two aspects of the analytic treatment of these ratings warrant clarification. First, the values entering the regression models are not individual ordinal judgments but *means across at least three independent expert raters*. An average of three or more ordinal judgments is a bounded, quasi-continuous quantity, and its use as a perceptual reference standard is standard practice in this literature: AVQI and ABI were themselves derived by multiple regression against mean perceptual ratings. Second, the principal association measure reported throughout is Spearman's r_s , which is rank-based and makes no interval-scale assumption, and the classification analyses require only ordinality. As a further safeguard, the CAPE-V analyses use a 100-point visual analog scale that is continuous by construction, while the GRBAS analyses use a four-point ordinal scale; the two converge on essentially the same parameter orderings (*Convergence of the Two Rating Systems*), which is direct evidence that the findings are not artifacts of scale properties.

Unsupervised redundancy analysis

All acoustic parameters were standardized to zero mean and unit variance. QR factorization with column pivoting was computed on the standardized design matrix $\mathbf{X}$ ($N \times 14$). The permutation order reveals which parameters contribute the most linearly independent information: the first pivoted column has the largest norm, the second has

the largest residual norm after projecting out the first, and so on. The diagonal elements $|R_{kk}|$ quantify each parameter's incremental contribution; a sharp decay indicates redundancy. Singular value decomposition (SVD) was performed on the same matrix for comparison, with the effective rank determined by a 99% cumulative energy threshold.

Supervised parameter ranking

For each perceptual target, Orthogonal Matching Pursuit (OMP) was used to rank the 14 parameters by their incremental predictive value. OMP is a greedy forward selection algorithm that iteratively: (1) identifies the parameter with the largest absolute Pearson correlation with the current residual, (2) adds it to the selected set, (3) refits ordinary least squares on all selected parameters, and (4) updates the residual.¹⁴ This produces a parameter ordering specific to each perceptual target, accounting for multivariate redundancy—a parameter that is highly correlated with a perceptual dimension but redundant with an already-selected parameter will be ranked lower than a moderately correlated but nonredundant one. Both predictors and target were standardized before analysis, and predictor columns were L2-normalized.

Bootstrap stability was assessed using 2000 resamples. For each resample, the full OMP ordering was computed, and the frequency with which each parameter appeared in the top three and top five positions was recorded.

Optimal number of parameters

For each perceptual target, the optimal number of parameters k was determined by evaluating models with $k = 1, 2, \dots, 10$ parameters (in OMP order) using three converging criteria:

1. **Adjusted R^2 :** The largest k for which adding the k -th parameter improved adjusted R^2 by at least 0.01.
2. **Bayesian Information Criterion (BIC):** The k yielding the minimum BIC, which penalizes model complexity more heavily than AIC.
3. **Cross-validated prediction error:** Tenfold cross-validation repeated 20 times, with the 1-SE rule selecting the simplest model within one standard error of the minimum cross-validated RMSE.

When all three criteria agreed, the common k was adopted. When they disagreed, the BIC-optimal k was used as the primary selection.

Classification performance against dichotomized perceptual ratings

Parameter selection was performed entirely on the continuous and ordinal perceptual ratings, as described above; no dichotomization enters the derivation of the subsets at any stage. The classification analysis reported here is a secondary presentation, included because AUC is the metric in which the performance of AVQI, ABI, and CSID is reported in the literature, and without it, no comparison

with those indices would be possible. It should also be emphasized that the reference standard is a dichotomized *perceptual rating*, not an independently established clinical diagnosis: the analysis quantifies agreement with expert auditory-perceptual judgment and does not establish diagnostic validity against a laryngoscopic or medical criterion.

For each dimension-specific index, ordinary least squares regression was fitted on the full sample, yielding predicted scores that served as a continuous index. Receiver operating characteristic (ROC) analysis was performed using the same methodology and thresholds as a companion validation study of established indices on the same database.¹⁸ CAPE-V ≥ 10 for clinically relevant deviation, GRBAS ≥ 0.5 for the midpoint between normal and minimal severity. The reference indices (AVQI, ABI, and CSID) were computed on the 284-speaker set. Both thresholds were adopted a priori and were not derived from the distribution of ratings in the present sample: CAPE-V ≥ 10 on the 100-point visual analog scale reflects the conventional criterion for clinically relevant deviation, and GRBAS ≥ 0.5 is the midpoint between “normal” (0) and “mild” (1), the natural ordinal analog. Critically, these are the same thresholds, applied to the same speakers, that were used to evaluate AVQI, ABI, CSID, and CPPS in the companion study¹⁸ from which the comparison values are drawn. The dichotomization is therefore identical across the dimension-specific subsets and the established indices, and the comparison is not confounded by differential thresholding. Area under the ROC curve (AUC) was computed as the exact nonparametric statistic, with bootstrap 95% confidence intervals from 2000 resamples and Youden’s J index²² for cutoff selection. Tenfold stratified cross-validation was repeated 100 times, with cutoffs derived on training folds and evaluated on held-out test folds.

Statistical Software

All analyses were performed in Python 3.11 using NumPy, SciPy,²³ pandas, scikit-learn,²⁴ and Matplotlib. Analysis code is available at <https://github.com/lucerojc/pvqd-qr-analysis>.

RESULTS

Sample characteristics

CAPE-V Overall Severity ratings ranged from 0.33 to 98.50 (mean = 28.63, SD = 24.55). GRBAS Grade is an ordinal scale, and mean and standard deviation are therefore not reported as descriptive summaries of its distribution; the ratings are instead summarized by median and interquartile range (median = 0.79, IQR = 0.33–1.67) and by the frequency distribution across the severity categories defined by the database authors (Table 3). The mean GRBAS value is used only as a regression target, where it represents an average across at least three independent raters rather than a single ordinal grade (*Perceptual Ratings*). The

TABLE 3.
Distribution of GRBAS Grade Across the Severity Categories Defined by the Database Authors¹⁵

GRBAS Grade	Normal (0-0.5)	Mild (0.6-1.5)	Moderate (1.6-2.5)	Severe (2.6-3.0)
<i>n</i> (%)	111 (39.1%)	100 (35.2%)	43 (15.1%)	30 (10.6%)

TABLE 4.
Descriptive Statistics for Acoustic Parameters

Parameter	<i>N</i>	Mean	SD	Min	Max
F0 (Hz)	288	174.62	62.56	75.85	413.77
log Jitter	288	− 0.35	0.83	− 1.76	2.24
log Shimmer (%)	288	1.32	0.74	− 0.07	3.17
log Shimmer (dB)	287	− 1.10	0.74	− 2.48	0.79
HNR (dB)	288	20.50	7.30	− 1.30	33.08
CPPS (dB)	288	13.18	4.23	2.34	22.42
Slope (dB)	285	− 21.91	5.00	− 35.24	− 4.71
Tilt (dB)	285	− 10.47	1.66	− 14.07	− 0.61
Alpha Ratio (dB)	288	− 12.45	5.03	− 29.53	4.83
GNE	288	0.85	0.13	0.42	0.99
Hno-6000	288	2.18	0.32	1.20	3.11
HNR-D (dB)	288	30.18	7.23	14.86	43.05
H1-H2 (dB)	288	4.20	6.14	− 20.62	25.15
log PSD	288	− 8.30	1.40	− 10.79	− 5.36

Notes: Jitter (%), shimmer (%), shimmer (dB), and PSD were log-transformed (natural logarithm) prior to analysis. Statistics shown are for the transformed values. Descriptive statistics use all available values per parameter; the multivariate analyses use the 284 complete cases.

distribution was skewed toward lower severity levels, consistent with the original database characteristics.¹⁵

Descriptive statistics for all 14 acoustic parameters (after log transformation of the four skewed parameters) are presented in Table 4. Three records had missing values in slope and tilt (due to very short vowel duration after pre-processing) and one had a missing value in shimmer (dB) (severely disordered voice where amplitude tracking failed), yielding 284 complete cases for multivariate analyses.

Acoustic parameter redundancy

The condition number of the standardized design matrix was 92.9, indicating moderate multicollinearity. SVD analysis revealed that 11 of 14 singular values were needed to capture 99% of the cumulative variance (Figure 1), indicating that approximately three of the 14 measured parameters carry information that is nearly entirely captured by the remaining 11.

QR factorization with column pivoting produced the parameter ordering shown in Table 5. F0 and H1-H2 were pivoted first (contributing the most linearly independent information), followed by GNE and Hno-6000. Slope was consistently the last parameter pivoted, with a relative diagonal contribution of only 0.038—indicating that it is

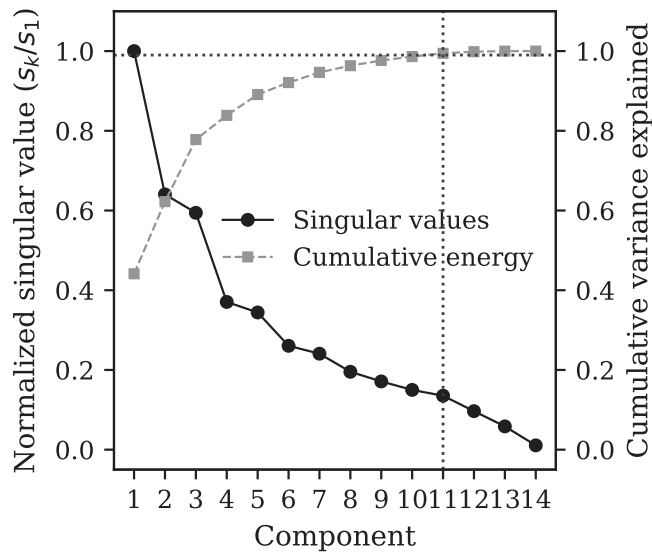


FIGURE 1. Singular value spectrum of the standardized acoustic parameter matrix ($N = 284$, $P = 14$). The dotted line indicates the 99% cumulative energy threshold, reached at $k = 11$.

TABLE 5.
Unsupervised QR Factorization: Parameter Ranking by Linear Independence ($N = 284$)

Rank	Parameter	$ R_{kk} $	$ R_{kk} /R_{11}$	Cumulative Energy
1	F0	16.82	1.000	0.152
2	H1-H2	16.81	0.999	0.303
3	GNE	16.64	0.989	0.452
4	Hno-6000	15.22	0.905	0.576
5	log PSD	13.62	0.810	0.675
6	Alpha Ratio	13.52	0.804	0.774
7	log Jitter	11.09	0.659	0.839
8	Tilt	9.18	0.546	0.885
9	log Shimmer (dB)	8.10	0.482	0.920
10	HNR-D	7.69	0.457	0.952
11	CPPS	7.54	0.448	0.982
12	HNR	4.70	0.279	0.994
13	log Shimmer (%)	3.32	0.197	1.000
14	Slope	0.64	0.038	1.000

almost entirely redundant with the other 13 parameters. Log shimmer (%) was also highly redundant (rank 13), largely captured by log shimmer (dB) and HNR.

Dimension-specific parameter rankings

OMP rankings differed substantially across perceptual dimensions (Table 6, Figure 2). Log shimmer (dB) was ranked first for overall severity (CAPE-V Severity and GRBAS Grade), GNE was ranked first for breathiness (CAPE-V and GRBAS), log shimmer (dB) was also ranked first for roughness, and HNR was ranked first for strain. The top three parameters for each dimension were:

- **Overall severity:** log shimmer (dB), GNE, and Hno-6000
- **Breathiness:** CPPS, GNE, Hno-6000 (CAPE-V) or GNE, Hno-6000, and CPPS (GRBAS)
- **Roughness:** log shimmer (dB), H1-H2, and GNE
- **Strain:** HNR, F0, and log PSD

CAPE-V and GRBAS scales produced identical parameter orderings for roughness and nearly identical orderings for severity (differing at rank 4: Hno-6000 for CAPE-V vs log PSD for GRBAS) and breathiness (CPPS first for CAPE-V, GNE first for GRBAS).

Bootstrap stability analysis confirmed the robustness of the top-ranked parameters (Table 7). For severity, GNE appeared in the top three in 71.9% of 2000 bootstrap resamples, with log shimmer (dB) in 42.3%; for breathiness, GNE appeared in the top three in 96.0%; for roughness, H1-H2 appeared in 71.5% and GNE in 67.9%. The severity rankings showed more variability than in the untransformed analysis, reflecting the competition between log shimmer (dB), GNE, and other noise measures for the top positions—a consequence of the log transformation redistributing predictive information more evenly among perturbation parameters. Rankings beyond position 3 showed greater variability, particularly for strain, consistent with the lower overall predictive accuracy of that dimension.

Optimal number of parameters

For each perceptual dimension, the three model selection criteria (adjusted R^2 , BIC, and cross-validated RMSE) converged on the same optimal k for five of seven targets (Figure 3). For the remaining two (GRBAS Grade and GRBAS Roughness), the criteria selected values of k differing by two (four vs six for Grade; four vs six for Roughness), with BIC as the primary criterion. The BIC-optimal selections are summarized in Table 8.

Overall severity required six parameters, breathiness three, roughness five, and strain six. For severity, roughness, and breathiness, the CAPE-V and GRBAS scales converged on the same top-ranked parameters, differing primarily in which parameters entered at lower ranks. The GRBAS Grade index notably included tilt (rank 4) in place of Hno-6000, reflecting the compressed ordinal scale's differential sensitivity to spectral shape.

Regression coefficients

Table 9 presents the ordinary least squares regression coefficients for the four CAPE-V indices. All coefficients were statistically significant ($P < 0.05$). The coefficient signs are consistent with known acoustic-perceptual relationships: parameters reflecting vocal fold regularity and harmonic structure (GNE, Hno-6000, HNR-D, HNR, and CPPS) have negative coefficients—higher values indicate healthier voice—while perturbation and instability measures (log shimmer dB, log PSD, and log jitter) have positive coefficients—higher values indicate greater

TABLE 6.
OMP Parameter Rankings by Perceptual Dimension. Lower Rank Indicates Greater Importance

Parameter	Sev	Breath	CAPE-V			GRBAS	
			Rough	Strain	Grade	Breath	Rough
log Shimmer (dB)	1	8	1	9	1	8	1
GNE	2	2	3	6	2	1	3
Hno-6000	3	3	11	12	7	2	7
log PSD	4	4	4	3	3	5	4
F0	5	5	9	2	5	6	5
HNR-D	6	7	6	7	6	7	10
log Jitter	7	12	5	10	9	13	6
Tilt	8	9	7	4	4	11	12
H1-H2	9	6	2	5	11	4	2
HNR	10	11	8	1	8	9	8
Slope	11	10	10	8	10	10	9
CPPS	12	1	12	11	12	3	11
Alpha Ratio	13	14	13	13	13	14	14
log Shimmer (%)	14	13	14	14	14	12	13

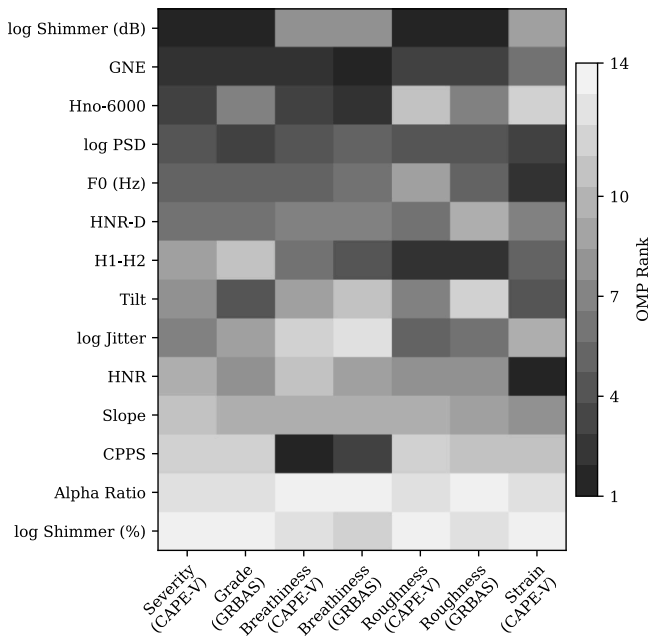


FIGURE 2. OMP parameter rankings across perceptual dimensions. Darker shading indicates higher importance (lower rank number). The ranking pattern shifts across dimensions: log shimmer (dB), GNE, and noise-related measures dominate for severity; CPPS and GNE dominate for breathiness; perturbation measures (log shimmer, H1-H2) dominate for roughness; and HNR and F0 dominate for strain.

dysphonia. F0 has a small positive coefficient in the severity and strain models, consistent with the elevated F0 often observed in effortful or tense phonation.

Classification performance against dichotomized perceptual ratings

ROC analysis results are presented in Table 10 and Figure 4. Cross-validated AUC estimates closely replicated the

resubstitution values (optimism ≤ 0.006 across all indices), confirming negligible overfitting.

The severity index (six parameters) achieved AUC = 0.870 for CAPE-V Severity, compared with AUC = 0.827 for AVQI rescored on the identical 284-speaker sample¹⁸ ($\Delta\text{AUC} = +0.043$). The breathiness index (three parameters) achieved AUC = 0.865, slightly exceeding the ABI's AUC of 0.859 ($\Delta\text{AUC} = +0.006$) with three rather than nine parameters. The roughness index (five parameters) achieved AUC = 0.830 and $r_s = 0.712$, improving on the strongest single parameter for the dimension (shimmer, $r_s = 0.68$). These comparisons are descriptive: the indices were derived from sustained vowels only, while AVQI and ABI used concatenated vowel and connected speech, and a formal paired comparison (DeLong test) would require computing both the data-driven indices and the established indices from the same signal type on the same subjects.

DISCUSSION

Summary of findings

This study applied rank-revealing matrix decomposition and greedy forward selection to identify optimal acoustic parameter subsets for each major perceptual voice quality dimension. Four principal findings emerged. First, the 14-parameter acoustic space contains substantial redundancy, with an effective dimensionality of approximately 11 and slope contributing negligible independent information. Second, log-transforming right-skewed perturbation parameters (jitter, shimmer, and PSD) improved linear prediction of severity incrementally ($\Delta r_s = +0.033$; $\Delta\text{AUC} = +0.024$) and of roughness substantially ($\Delta r_s = +0.069$; $\Delta\text{AUC} = +0.049$), with log shimmer (dB) displacing CPPS as the most important predictor of overall severity. Third, the optimal parameter subset differs markedly across perceptual dimensions, with severity dominated by log

TABLE 7.

Bootstrap Stability of OMP Rankings (2000 Resamples): Percentage of Resamples in Which Each Parameter Appeared in the Top-3 Positions. Only Parameters with $\geq 20\%$ Frequency Shown

Parameter	CAPE-V Severity	CAPE-V Breathiness	CAPE-V Roughness	CAPE-V Strain
log Shimmer (dB)	42.3		48.4	37.8
GNE	71.9	96.0	67.9	
Hno-6000	37.0	60.4		
log PSD	36.9		30.0	38.2
CPPS	26.7	57.3		
H1-H2			71.5	54.3
HNR				46.4
F0				49.2
Tilt	28.1	25.1		41.9
log Jitter			26.6	
log Shimmer (%)			25.2	

Notes: Empty cells indicate $< 20\%$ frequency.

shimmer (dB) and noise measures, breathiness dominated by CPPS and noise measures, roughness dominated by perturbation and source measures, and strain uniquely requiring F0 and spectral tilt. Fourth, using sustained vowels alone, dimension-specific subsets match or exceed the classification performance of established multiparametric indices scored on connected speech as well as the vowel—a task-asymmetric comparison that favors the established

indices, and one that does not establish superiority in their intended multitask use.

Dimension-specific parameter importance

The finding that different perceptual dimensions require different acoustic parameter subsets has direct implications for clinical assessment and index design. The severity index is dominated by log shimmer (dB), GNE, Hno-6000, and

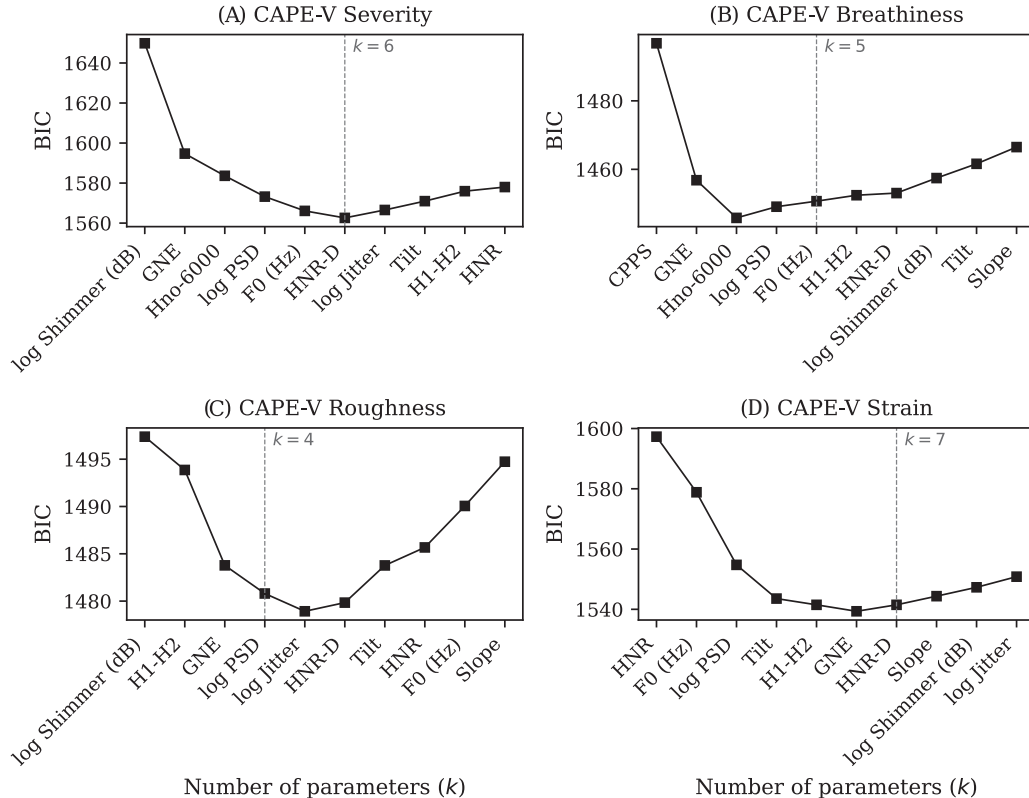


FIGURE 3. Bayesian Information Criterion (BIC) as a function of the number of parameters k (added in OMP order) for each CAPE-V dimension. Dashed vertical lines indicate the BIC-optimal k . Lower BIC indicates better model fit after penalizing complexity. The optimal k ranged from 3 (breathiness) to 6 (severity, strain).

TABLE 8.
BIC-Optimal Indices for Each Perceptual Dimension

Target	<i>k</i>	Parameters (in OMP Order)	Adj R^2	RMSE	r_s
CAPE-V Severity	6	log Shim. dB, GNE, Hno-6000, log PSD, F0, and HNR-D	0.637	15.06	0.772
CAPE-V Breathiness	3	CPPS, GNE, and Hno-6000	0.616	12.46	0.727
CAPE-V Roughness	5	log Shim. dB, H1-H2, GNE, log PSD, and log Jitter	0.500	13.10	0.712
CAPE-V Strain	6	HNR, F0, log PSD, Tilt, H1-H2, and GNE	0.529	14.55	0.657
GRBAS Grade	6	log Shim. dB, GNE, log PSD, Tilt, F0, and HNR-D	0.624	0.55	0.773
GRBAS Breathiness	3	GNE, Hno-6000, and CPPS	0.590	0.48	0.712
GRBAS Roughness	4	log Shim. dB, H1-H2, GNE, and log PSD	0.508	0.47	0.700

Notes: *k* = number of parameters; RMSE = root mean square error (in CAPE-V units or GRBAS units); r_s = Spearman correlation between predicted and observed ratings; Adj R^2 = adjusted coefficient of determination. *N* = 284 complete cases. log = natural logarithm.

log PSD—a combination of perturbation and noise measures that collectively characterize the degradation of periodic vocal fold vibration. The breathiness index shares two of these parameters (GNE, Hno-6000) but is led by CPPS, reflecting the distinct acoustic signature of incomplete glottal closure. In contrast, the roughness index is dominated by log shimmer (dB), H1-H2, and log jitter—parameters that capture cycle-to-cycle irregularity—while the strain index uniquely prioritizes HNR, F0, and spectral tilt, reflecting the effortful, pressed phonation characteristic of strained voice quality.

Perhaps the most striking finding is the displacement of CPPS by log shimmer (dB) as the top predictor of overall severity. In the untransformed analysis, CPPS ranked first, consistent with the large body of literature identifying

CPPS as the single best acoustic predictor of dysphonia.^{25,1}

However, after log-transforming the skewed perturbation parameters, log shimmer (dB) overtook CPPS—which dropped to rank 12—while severity prediction improved substantially (r_s from 0.74 to 0.77, AUC from 0.846 to 0.870). This suggests that the widely reported primacy of CPPS may partly reflect a distributional advantage: CPPS is computed via cepstral analysis (itself a log-spectral operation), giving it an approximately symmetric distribution that is well-suited to linear regression, whereas raw shimmer and jitter have strongly right-skewed distributions that compress discriminative information into a narrow range. Once this distributional asymmetry is removed by log transformation, perturbation measures emerge as more informative predictors of overall severity.

TABLE 9.
Regression Coefficients for CAPE-V Dimension-Specific Indices

Index	Predictor	β	SE	<i>t</i>	<i>P</i>
Severity (<i>k</i> = 6)	Intercept	147.66	11.26	13.11	< 0.001
	log Shimmer (dB)	6.18	2.11	2.93	0.004
	GNE	− 55.84	8.89	− 6.28	< 0.001
	Hno-6000	− 13.27	3.32	− 4.00	< 0.001
	log PSD	3.64	0.89	4.06	< 0.001
	F0	0.09	0.02	4.68	< 0.001
	HNR-D	− 0.67	0.22	− 3.01	0.003
Breathiness (<i>k</i> = 3)	Intercept	117.00	6.90	16.96	< 0.001
	CPPS	− 1.60	0.29	− 5.60	< 0.001
	GNE	− 61.87	8.80	− 7.03	< 0.001
	Hno-6000	− 10.91	2.65	− 4.11	< 0.001
Roughness (<i>k</i> = 5)	Intercept	67.95	8.50	7.99	< 0.001
	log Shimmer (dB)	6.24	2.04	3.05	0.003
	H1-H2	− 0.43	0.14	− 3.18	0.002
	GNE	− 27.83	7.35	− 3.79	< 0.001
	log PSD	1.60	0.78	2.06	0.041
	log Jitter	4.85	1.77	2.73	0.007
	Intercept	110.92	9.77	11.36	< 0.001
Strain (<i>k</i> = 6)	HNR	− 0.84	0.18	− 4.70	< 0.001
	F0	0.07	0.02	4.27	< 0.001
	log PSD	4.49	0.83	5.41	< 0.001
	Tilt	2.30	0.59	3.92	< 0.001
	H1-H2	− 0.52	0.15	− 3.37	< 0.001
	GNE	− 24.55	8.84	− 2.78	0.006

TABLE 10.
Classification Performance of Dimension-Specific Indices Against Dichotomized Perceptual Ratings

Index	Reference Standard	Resubstitution				Cross-Validation		
		AUC	Cutoff	Sens	Spec	AUC	Sens	Spec
Severity	CAPE-V Sev ≥ 10	0.870	18.34	79.3%	81.1%	0.867	78.1%	77.7%
Severity	GRBAS G ≥ 0.5	0.872	0.78	75.7%	86.4%	0.869	74.8%	83.5%
Breathiness	CAPE-V B ≥ 10	0.865	12.33	84.3%	72.9%	0.863	77.7%	74.7%
Breathiness	GRBAS B ≥ 0.5	0.843	0.48	78.2%	76.0%	0.841	76.6%	72.8%
Roughness	CAPE-V R ≥ 10	0.830	16.24	74.4%	79.4%	0.825	75.5%	73.5%
Roughness	GRBAS R ≥ 0.5	0.832	0.54	83.0%	69.4%	0.828	78.3%	68.4%
Strain	CAPE-V S ≥ 10	0.790	19.90	65.0%	85.6%	0.784	63.3%	83.5%

Notes: Resubstitution: cutoff derived and evaluated on the full sample using Youden's J index. Cross-validation: 10-fold stratified, repeated 100 times. $N = 284$ (complete cases) for all dimensions. Cutoff values are from resubstitution and should be considered hypothesis-generating; external validation is required before clinical use.

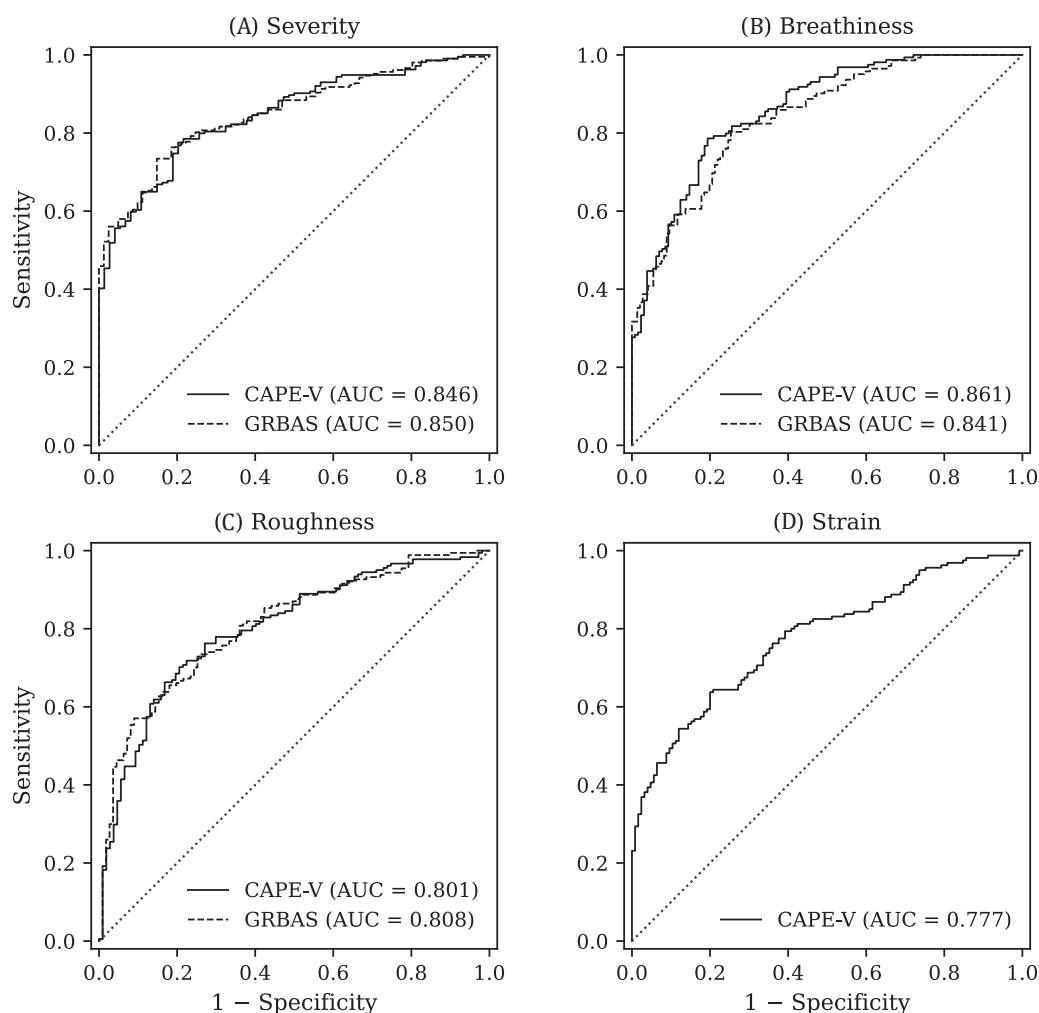


FIGURE 4. ROC curves for dimension-specific indices. Each panel shows the ROC curve for one index applied to its target perceptual dimension(s). AUC values with bootstrap 95% confidence intervals are shown.

Two clarifications are important, since the exclusion of CPPS from the severity subset might otherwise appear difficult to reconcile with the literature. First, CPPS was *not* excluded because of a weak association with severity, nor because of collinearity with another selected parameter, nor

as an artifact of the selection procedure. It was *displaced*: in the untransformed analysis, it ranked first, and it fell in rank only after the four right-skewed perturbation parameters were placed on a scale comparable to its own. The mechanism is distributional, and the resulting claim is

correspondingly narrow—that part of the long-reported primacy of CPPS in *linear* models may reflect an advantage conferred by its implicit log-scaling, an advantage that competing perturbation measures do not receive unless it is granted to them explicitly. This is a testable proposition rather than an established fact, and it is offered as such. Second, CPPS remains the top-ranked parameter for CAPE-V breathiness and is retained in the breathiness subset (Table 8). It is displaced only for overall severity, and only after transformation; nothing in these results argues against CPPS as a robust acoustic correlate of dysphonia in general.

The relevance of this transformation is not hypothetical, and there is a direct precedent in the development of ABI. Barsties v. Latoszek et al.⁴ computed both the raw period standard deviation (PSD) and its natural logarithm (LNPSD) among their 28 candidate measures, explicitly noting that the logarithm was applied “to correct the feature of nonlinearity in PSD.” Both forms showed the weakest bivariate association with perceived breathiness of any parameter in their set ($r_s = -0.069$ for each), and the authors report that although LNPSD had been identified as an acoustic predictor of breathiness, *the nonlogarithmic form was the one implemented in ABI*. The present results indicate that this choice may have been consequential: on the log scale, PSD is retained in six of the seven dimension-specific subsets derived here, and log shimmer (dB) displaces CPPS as the leading predictor of overall severity. A measure that appears negligible in bivariate terms on a raw scale can carry substantial predictive weight in a multivariate model once its distribution is corrected—which is precisely the pattern reported here.

For direct comparison with studies using CPPS alone, CPPS from sustained vowels on this same database and sample yields AUC = 0.796 for CAPE-V Overall Severity, against AUC = 0.870 for the six-parameter severity subset.¹⁸ This comparison is cleaner than that against AVQI, ABI, and CSID, because CPPS is evaluated on the same speech task, in the same speakers, and under the same threshold; the incremental value of the multiparametric subset over CPPS alone is therefore not confounded by task differences.

Physiological interpretation of selected parameters

The parameter selections can be interpreted in terms of the underlying voice production mechanisms for each perceptual dimension.

Breathiness arises from incomplete glottal closure during phonation, allowing turbulent airflow to escape through the glottal gap. The three top-ranked parameters for breathiness—GNE, Hno-6000, and CPPS—each capture a different acoustic consequence of this mechanism. GNE directly quantifies the ratio of periodic (harmonic) to aperiodic (noise) excitation at the glottis.²⁶ Hno-6000 captures the relative energy in the high-frequency noise band (above 6 kHz), where aspiration noise from glottal air leakage is most prominent. CPPS reflects the overall loss of

harmonic periodicity in the signal, which decreases as turbulent noise fills the inter-harmonic spectral valleys.

Roughness is associated with irregular vocal fold vibration—asymmetric oscillation, aperiodic cycle-to-cycle variation, or subharmonic patterns. Shimmer, ranked first for roughness, directly reflects cycle-to-cycle amplitude instability caused by irregular fold contact. H1-H2 (ranked second) reflects the spectral slope of the glottal source: irregular vibration produces a flatter spectral envelope (reduced H1-H2 difference), consistent with the negative coefficient observed ($\beta = -0.43$). The relatively lower predictive accuracy for roughness (Adj $R^2 = 0.500$) compared with breathiness (Adj $R^2 = 0.616$) may reflect the fact that roughness encompasses a heterogeneous set of vibratory abnormalities—from mild cycle-to-cycle irregularity to diplophonia and subharmonic breaks—that may not be fully captured by linear measures from sustained vowels.

Strain reflects effortful, hyperfunctional phonation with increased laryngeal muscle tension. The positive F0 coefficient ($\beta = 0.07$) is consistent with increased longitudinal tension of the vocal folds (cricothyroid muscle activation) that elevates fundamental frequency. The positive tilt coefficient ($\beta = 2.30$) indicates that less negative spectral tilt—ie, relatively more high-frequency energy—predicts strain, consistent with the increased medial compression and firm glottal closure that characterize pressed voice quality.²⁷ The fact that strain requires among the largest parameter sets ($k = 6$, equal to overall severity) and achieves the lowest accuracy (AUC = 0.790) suggests that it is a perceptually complex dimension encompassing multiple production mechanisms that are only partially captured by acoustic analysis of sustained vowels.

Task dependence and ecological validity

The exclusive use of sustained vowels was a deliberate design choice—it isolates voice source characteristics from articulatory, prosodic, and linguistic influence, and yields language-independent measurements—but it bounds the interpretation of these results in three ways that deserve explicit statement.

First, the sustained vowel is a reduced-complexity task, carrying no articulatory movement, no prosodic variation, no phonetic context, and no speech-planning demand. This controlled phonatory context plausibly accounts for part of the predictive accuracy reported here, and the values obtained should be regarded as an *upper bound* on what would be expected in connected speech. Connected speech, while more variable, carries clinically meaningful information about voice quality—vocal onset and offset behavior, phonatory stability under articulatory load, and voice breaks in running speech—that sustained vowels cannot express and that perceptual raters demonstrably use.

Second, the elevated ranking of perturbation measures is conditional on this task. Jitter, shimmer, and PSD depend on reliable cycle-to-cycle period detection, which is comparatively well-behaved in a sustained vowel and considerably less so in connected speech, where voicing

boundaries, phonetic variation, and prosodic movement degrade F_0 tracking; their known unreliability in moderate-to-severe dysphonia^{28,29} compounds this. No claim is made—and none should be inferred—that perturbation measures are broadly robust markers of voice quality across speech tasks. Whether the log transformation confers a comparable benefit in connected speech is an open empirical question and a specific target for future work.

Third, and consequently, the parameter subsets derived here are optimal *for the sustained-vowel task specifically*. Their composition should not be assumed to transfer to connected speech, and the comparison with the established indices that follows must be read in this light.

Comparison with established indices

The comparison presented in this section is task-asymmetric, and the asymmetry runs against the present study rather than in its favor. AVQI and ABI were scored on the concatenation of connected speech and the sustained vowel, and CSID on connected speech, as they were designed to be; the dimension-specific subsets derived here use the sustained vowel alone. The reference indices therefore had access to *more* speech material, not less. Two consequences follow. Any advantage shown by the subsets is obtained despite this handicap, and is best understood as a claim about parameter selection and parsimony rather than about the clinical validity of AVQI or ABI in their intended multitask use. Conversely, where a subset falls short of a reference index, the shortfall may reflect the missing connected speech information rather than a deficiency of the selected parameters. A properly matched comparison—vowel-only subsets against vowel-only indices—remains to be performed, and no claim of superiority in the indices' intended context is made here.

The severity index achieved AUC = 0.870 compared with AVQI's 0.827, using the same number of parameters (6) but a substantially different composition. The index relies on log-transformed perturbation (log shimmer dB, log PSD), noise-related measures (GNE, Hno-6000, and HNR-D), and source characteristics (F0); notably excluding CPPS, the parameter most commonly recommended for dysphonia screening. This suggests that once perturbation measures are placed on a logarithmic scale, they carry predictive information about overall severity that is not recoverable from cepstral and noise measures alone—a reversal of the pattern seen in the untransformed analysis, where CPPS ranked first and perturbation measures contributed little incremental value.

The breathiness index slightly exceeded ABI's classification performance (AUC = 0.865 vs 0.859, both resubstitution estimates on the identical 284-speaker sample) with three parameters instead of nine, suggesting that several ABI components may be partially redundant. However, these comparisons should be interpreted cautiously. First, the data-driven indices were derived from and evaluated on the same database, and despite cross-validation, the risk of overoptimistic estimates remains; the

established indices (AVQI, ABI) were developed on proprietary clinical samples and have been externally validated across multiple languages and populations.^{30,31} Second, the data-driven indices used sustained vowels only, while AVQI and ABI were computed from concatenated vowel and connected speech signals. Third, unlike the AVQI and ABI—which were developed on proprietary clinical samples—the present analysis used a publicly available database, enabling full reproducibility of the parameter selection process; however, this same feature means the results require external validation before they can be considered alternatives to the established measures.

The CSID⁵ is built on two acoustic measures—cepstral peak prominence (CPP or CPPS) and a low-to-high-frequency spectral energy ratio (L/H)—which, together with their variability, enter the regression as four terms. L/H is not among the 14 parameters analyzed here and is not equivalent to the alpha ratio, which uses different frequency bands. The alpha ratio nonetheless provides a related indicator of low-vs-high spectral balance, and its behavior in the present analysis is informative. Alpha ratio was among the least informative parameters in every analysis conducted in this study: rank 13 in the supervised OMP orderings for severity and rank 13–14 across all dimensions. Its near-complete redundancy with the other 13 parameters in the unsupervised QR factorization (slope, rank 14, relative $|R_{kk}| = 0.038$) indicates that the spectral tilt information it captures is already present in other measures—particularly tilt and CPPS. Whether the same redundancy holds for L/H, computed over different bands, cannot be established from these data. What can be said empirically is that the severity index derived here—combining noise-related measures (GNE, Hno-6000, and HNR-D) with log-transformed perturbation measures—achieves substantially better classification performance than CSID rescored on the same 284-speaker sample (AUC = 0.870 vs 0.787).¹⁸

Novel indices for roughness and strain

No published composite acoustic index specifically targets roughness or strain. The roughness index (log shimmer dB, H1-H2, GNE, log PSD, and log jitter) achieved AUC = 0.830 (CAPE-V; 0.832 for GRBAS) with five parameters, providing an initial candidate for a dimension-specific roughness measure. The log transformation was particularly beneficial for roughness, improving r_s from 0.64 to 0.71 and AUC from 0.781 to 0.830—the largest gain among all dimensions. The strain index achieved AUC = 0.790 with six parameters, representing the first data-driven acoustic characterization of this perceptual dimension. Both should be considered hypothesis-generating, external validation is essential before clinical adoption.

Convergence of the two rating systems

Both CAPE-V and GRBAS were retained deliberately, and they are not treated as independent evidence: their results are reported in parallel and are never pooled. The two

scales were rated by the same clinicians on the same speakers, but differ in a way that is analytically useful—CAPE-V is a continuous 100-point visual analog scale, GRBAS a four-point ordinal scale. If the parameter rankings reported here were artifacts of scale properties, whether of visual-analog anchoring or of the averaging of ordinal judgments, the two scales would be expected to yield divergent orderings. They do not: the orderings are identical for roughness and differ by a single rank position for severity and for breathiness. Overlapping dimensions are reported side by side precisely so that this agreement is visible and auditable. The convergence is presented as a robustness check on the parameter rankings, and it speaks directly to the concern that averaged ordinal ratings might not support the analyses applied to them.

Methodological considerations

OMP was chosen over alternative feature selection methods for several reasons. Unlike PCA, which creates rotated components that combine multiple parameters, OMP ranks the original parameters directly—preserving clinical interpretability. Unlike LASSO or elastic net, OMP is deterministic (no regularization hyperparameter to tune) and produces a strict parameter ordering rather than a set of nonzero coefficients that varies with the penalty strength. Unlike p -value-based stepwise regression (used to develop AVQI and ABI), OMP operates on standardized and normalized predictors, making it less sensitive to distributional properties of individual parameters. The relationship between OMP and stepwise regression is close—both are greedy forward selection procedures—but OMP's use of residual correlations rather than partial F tests provides a more natural measure of incremental predictive value in the presence of collinearity.

The decision to log-transform jitter, shimmer, and PSD was motivated by their distributional properties (strong right skew, coefficient of variation > 0.9) rather than by outcome-dependent optimization. All four transformed parameters share the property of being strictly positive perturbation or instability measures whose pathological values span orders of magnitude. The transformation was applied uniformly to all four candidates before any model selection, ensuring that the OMP rankings and BIC selections were not biased by selective transformation. The resulting improvement—particularly the displacement of CPPS by log shimmer (dB) as the top severity predictor—suggests that established indices using untransformed perturbation parameters may underestimate the predictive information in these measures.

The bootstrap stability analysis (Table 7) confirmed that the reported rankings are robust: the top-ranked parameters for breathiness, roughness, and strain each appeared in the top 3 in $> 45\%$ of 2000 resamples. The severity rankings showed more variability than other dimensions, with GNE (71.9%) rather than log shimmer (dB) (42.3%) being the most stable top three parameter, despite

log shimmer (dB) ranking first in the point estimate. This variability reflects genuine competition among several strong predictors rather than instability of the method.

Clinical application

In clinical practice, the dimension-specific parameter selection can guide the design of targeted assessment protocols. A clinician screening for overall dysphonia severity needs log shimmer (dB), GNE, Hno-6000, log PSD, F0, and HNR-D—all computable from a single sustained vowel. A clinician monitoring breathiness (eg, in vocal fold paralysis) needs only three of those parameters (CPPS, GNE, Hno-6000). A clinician tracking roughness (eg, in vocal fold lesions) needs a different subset centered on log shimmer (dB), H1-H2, and log jitter. This dimension-specific approach avoids the “one-size-fits-all” limitation of existing indices, which were designed for a single perceptual dimension and may not optimally track others.

The log transformation adds minimal computational burden (a single operation per parameter) while substantially improving accuracy. For implementation, the index formulas can be applied directly to log-transformed values; clinicians need not interpret the log-transformed parameters themselves, only the resulting index scores.

For telehealth and smartphone-based assessment, where computational resources and recording quality may be limited, knowing that 4–6 parameters suffice for clinically useful accuracy provides practical guidance for application design. The language independence of sustained vowel parameters is an additional advantage in multilingual clinical contexts.

Limitations

Several limitations should be considered. First, the indices were derived and evaluated on the same database (PVQD), and despite cross-validation showing negligible AUC optimism (≤ 0.006), external validation on an independent sample is essential before clinical adoption. Such validation would require a database with both sustained-vowel recordings and expert perceptual ratings across all CAPE-V and GRBAS dimensions—a resource that, to the author's knowledge, does not currently exist publicly beyond the PVQD. Second, only sustained-vowel parameters were analyzed; connected speech parameters (which contribute to AVQI, ABI, and CSID) were deliberately excluded to isolate voice source characteristics from articulatory and linguistic effects, but this limits direct comparison with the established indices. Third, the linear (OLS) framework may not capture nonlinear acoustic-perceptual relationships beyond those addressed by the log transformation. Fourth, the OMP algorithm is greedy and may not identify the globally optimal parameter subset, though the bootstrap stability analysis showed consistent rankings across 2000 resamples. Fifth, the PVQD sample is skewed toward lower severity levels, which may affect the generalizability of the derived cutoffs. Sixth, the comparison with AVQI and ABI is descriptive rather than statistical (formal DeLong tests

require paired scores from both indices on the same subjects computed from the same signal type). Seventh, the choice of which parameters to log-transform was based on distributional properties (right skew, strict positivity) rather than on an exhaustive search of all possible transformation combinations; other transformations (eg, Box-Cox) might yield further improvements.

Five further limitations should be noted. Eighth, the reference standard is a perceptual rating, not an independently established clinical diagnosis; the analyses quantify agreement with expert auditory judgment, and high agreement with perceptual ratings does not by itself establish diagnostic validity against a laryngoscopic or medical criterion. Ninth, the sustained-vowel task is of reduced complexity relative to connected speech, and the accuracy reported here may partly reflect that reduction rather than the strength of the selected parameters alone (*Task Dependence and Ecological Validity*). Tenth, the classification analyses dichotomize the perceptual ratings, which simplifies the task, particularly given that the PVQD is skewed toward lower severity, so the classifier is to a considerable extent separating clearly normal voices from voices with any perceptual deviation. This inflation applies equally to the AUC values reported for AVQI, ABI, and CSID under identical thresholds, so it does not differentially favor the subsets derived here; the continuous analyses (r_s , adjusted R^2), which are primary, are unaffected by it. Eleventh, the comparison with the established indices is task-asymmetric in their favor, as discussed above, and a matched vowel-only comparison remains to be performed.

Twelfth, the accuracy attainable by any acoustic model is bounded above by the reliability of the perceptual reference standard, and the reported reliabilities (Table 2) qualify the results in an instructive way. Overall severity is both the most reliably rated dimension (ICC = 0.918) and the best predicted; roughness is the least reliably rated (ICC = 0.789 for CAPE-V, 0.787 for GRBAS) and yields the lowest adjusted R^2 , so part of its residual variance is plausibly attributable to a noisier target rather than to the acoustic parameters. Strain, however, does not fit this pattern, and the discrepancy is informative. Strain is rated as reliably as breathiness (ICC = 0.829 vs 0.827) and more reliably than roughness, yet it is the least well-predicted dimension on every metric (adjusted R^2 = 0.529; r_s = 0.66; AUC = 0.790). Its weak prediction, therefore, cannot be explained by rater disagreement: the listeners agree on what they hear, and the sustained-vowel parameters nonetheless fail to capture it. This points to a limitation of the speech material rather than of the reference standard. Strain reflects effortful, hyperfunctional phonation, and its perceptual cues may reside substantially in properties a sustained vowel cannot express—vocal onset behavior, phonatory effort sustained under articulatory load, and pitch and loudness variation in running speech. This is arguably the clearest evidence in the present data that dimension-specific acoustic modeling of strain will require connected speech (*Task Dependence and Ecological Validity*).

Future directions

External validation of the dimension-specific indices in independent samples and languages is the most critical next step. Extension to connected speech parameters would enable formal paired-sample comparisons with AVQI and ABI. The finding that log transformation substantially improves prediction warrants investigation of whether similar transformations would benefit existing indices (AVQI, ABI) that use untransformed perturbation parameters. Evaluation with smartphone-recorded audio would test ecological validity for telehealth applications. Finally, prospective evaluation of sensitivity to change following treatment would establish clinical utility for longitudinal monitoring.

CONCLUSIONS

Different perceptual dimensions of voice quality are optimally predicted by distinct subsets of acoustic parameters, with the number of required parameters ranging from three (breathiness) to six (severity, strain). Log-transforming right-skewed perturbation measures (jitter, shimmer, and PSD) improves linear prediction incrementally for overall severity, substantially for roughness, with log shimmer (dB) emerging as the single most important predictor of overall dysphonia severity, displacing CPPS, which has been widely recommended as the primary acoustic measure of dysphonia. Using sustained vowels alone, data-driven dimension-specific subsets match or exceed the classification performance of AVQI and ABI, which were scored on connected speech in addition to the vowel; because the reference indices had access to more speech material, this is not a claim of superiority in their intended use, but it does indicate that comparable performance is attainable from a single vowel and a small number of parameters. The analysis also provides the first empirically derived parameter sets for roughness and strain. These findings support the development of targeted, dimension-specific acoustic indices and indicate that simple parameter transformations can yield meaningful improvements in voice quality prediction, subject to external validation and confirmation in connected speech.

Data and Code Availability

The PVQD is publicly available at <https://data.mendeley.com/datasets/9dz247gnyb>.¹⁵ Acoustic parameter extraction used Parselmouth version 0.4.7²⁰ (Praat 6.1.38). Analysis code is available at <https://github.com/lucerojc/pvqd-qv-analysis>.

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Declaration of Generative AI and AI-Assisted Technologies in the Paper Preparation Process

During the preparation of this work, the author used Claude (Anthropic) to assist with statistical code development, paper drafting, and editorial revision. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

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